



First Name _____

This paper is for St Lucia Campus students.

Total

Problem 1 – Maths for Computer Science

Briefly describe the proof of Gödel's first incompleteness theorem ensuring that you:

- a) Explain how integers are used in the proof, type of mathematical proof that was used and the key mathematical mechanism that enables the proof to work.
(3 Marks)
- b) Explain how Alan Turing was inspired by Gödel's proof and what proof he constructed in the context of the theory of computation.
(2 Marks)

Problem 2 – Turing Machines

Describe a universal Turing machine using an example in a modern-day context and explain how Alan Turing used it to prove that the Halting problem is undecidable with its relation to the decision problem in mathematics (sometimes referred to as the 'Entscheidungsproblem').

(5 Marks)

Problem 3 – Turing Machines

Write down the formal description for the Turing machine that decides the language $L = \{s\#s \mid s \in \{0,1\}^*\}$. You may use a transition table or diagram to describe the transition rules for the machine.

(5 Marks)

Problem 4 – Lambda Calculus

Describe how Boolean logic can be defined in Lambda Calculus and use it to define the ISZERO combinator making sure to explain all the relevant arguments and function bodies. Show how it performs when it is applied to the number zero.
(5 Marks)

Question 5 – Lambda Calculus

Describe how one can compute the factorial function using recursion in Lambda Calculus. Make sure to mention all the relevant combinators and theorems required to achieve your result.

(7 Marks)

Problem 6 – Quantum Computation

Give an example of an important problem solvable by a quantum computer with its unique computation properties using the principles of quantum mechanics, and discuss why a quantum computer can solve problems much faster than a conventional computer.

(8 Marks)

Question 7 – Applied Problem

The Australian Space Agency (ASA) in collaboration with other space agencies around the world are commissioning a space probe mission to explore nearby star systems for life. They wish to include a set of instructions and a message that demonstrates human capability and our understanding of the universe within the spacecraft. The agencies have decided that computation will be the topic that best demonstrates this to any potential aliens.

You have been chosen by the top scientists in the ASA to design the message that will convey our understanding of computation to any potential aliens that might discover the probe. The message must meet the following requirements:

1. It must be readable by potential aliens without without any prior knowledge of the symbols beforehand.
2. It must be as simple and compact as possible as there is very little space for the message itself within the probe.

Your task is to use the theories of computation covered in this course to construct this message by:

- a) Choosing the model of computation (either Turing machines, Lambda Calculus or Quantum Computation) and justify why this model will convey our understanding of computation. (2 Marks)
- b) Choose and write down the necessary equations or expressions from your chosen model that will convey our understanding of computation. (3 Marks)
- c) Justify why these equations or expressions represent computation in its simplest and most compact form, making sure to highlight any important theorems or mathematical concepts of note. (5 Marks)
- d) Explain how all the information above will convey our thorough understanding of the universe to the aliens, making sure to discuss important aspects of computation in general. (5 Marks)

END OF EXAMINATION